TEXORA · LECTURE SHEET

Measurement of Yarn Hairiness

Hairs are the fibre ends and loops that escape the yarn body. A few give cotton its soft face; too many mean pilling, misty prints, warp breaks and wasted fibre. This sheet defines hairiness, shows how the photoelectric instruments count it, and explains what drives the number up or down.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 14 · LEVEL 3 TERM 1

1. What hairiness is

Yarn hairiness is the population of fibre ends and loops protruding from the yarn's compact body:

- **Protruding ends** — one end anchored in the yarn, the other free; formed by fibre ends that were never fully bound in, especially short fibres.
- **Loops** — both ends anchored, the middle bowed out; formed by fibres buckled during spinning or winding.
- **Wild fibres** — barely attached, lying along the surface; the first to become fly and lint.

Effects — good and bad

MODERATE HAIRINESS HELPS	EXCESS HAIRINESS HARMS
Soft handle and warm, covered fabric surface (brushed and knitted goods)	Pilling — hairs entangle into pills on the fabric face
Blurred, gentle print edges where wanted	Unsharp prints and dye specks; misty fabric appearance
—	Warp ends cling to each other in weaving → sheds stick, ends break
—	More fly and dust in the mill; fibre literally spun into the air

2. The measuring principle — photoelectric counting

The standard hairiness testers (the hairiness module of the evenness tester family, and dedicated instruments of the Zweigle type) pass the yarn at constant speed through a focused light beam / sensor

field. Each hair crossing the measuring zone scatters or interrupts light; the electronics convert the events into one of two report forms:

- **Hairiness index H** — the total length of protruding fibres, in centimetres, per centimetre of yarn, integrated over a measuring field extending about 1 cm from the yarn surface. H is a single comparable number per yarn.
- **Hair counts by length class** — the number of hairs per unit yarn length in each length band (e.g. 1, 2, 3, 4, 6, 8, 10, 12, 15, 18, 21, 25 mm classes in Zweigle-type reporting). From these, **S3** is computed: the number of hairs of length ≥ 3 mm per 100 m (in some conventions per stated test length) — the long hairs that drive pilling and weaving trouble, watched separately from the harmless short fuzz.

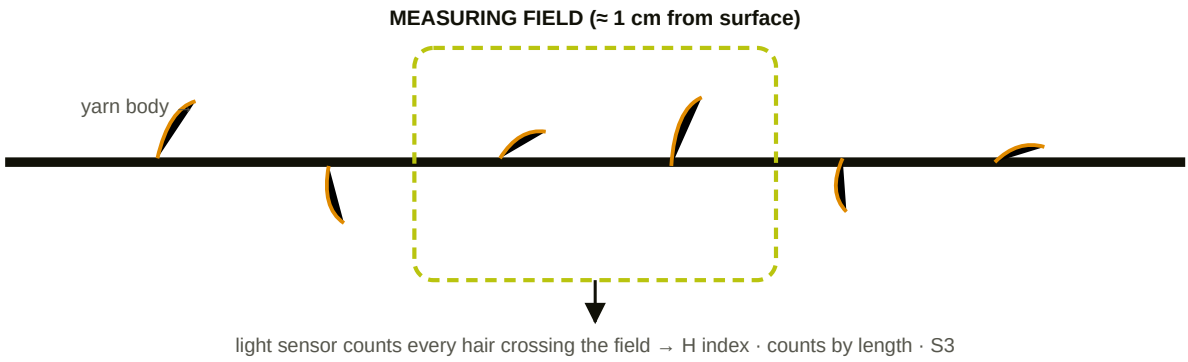


Figure 1 — Hairiness measurement principle. The yarn runs through a defined optical field; protruding ends and loops interrupt the light and are totalled as length (H) and as counts in length classes.

WORKED EXAMPLE — THE H INDEX AND A WINDING EFFECT

(a) A yarn tests at H = 5.2. Meaning: over each 1 cm of yarn there are, in total, **5.2 cm** of protruding hair within the 1 cm field — e.g. a 100 m test length carries about 5.2 × 10,000 = 52,000 cm = **520 m of hair**. Typical ring-spun carded cotton yarns read H ≈ 4.5–6.5 by count; combed and compact-spun yarns read markedly lower.

(b) The same yarn re-tested after high-speed winding reads H = 6.1. Winding — guide friction, tension peaks, package abrasion — has raised hairiness by (6.1 − 5.2)/5.2 × 100 = **17 %**. Hairiness specifications must therefore state the stage tested: spinning cop, cone after winding, or after singeing are three different numbers for one yarn.

3. What controls hairiness

- **Fibre length and short fibre content** (Topic 3) — the strongest material lever: every short fibre contributes two ends near the surface. Combing (removing short fibre as noil) is the classic hairiness cure.

- **Twist** (Topic 11) — higher twist binds ends in; hairiness falls as TM rises within the normal range.
- **Spinning system** — compact spinning (condensing the strand before twist insertion) gives the lowest hairiness of the staple systems; ring spinning is the reference; rotor yarns show a different structure with wrapper fibres; air-jet yarns are low in long hairs.
- **Machine condition** — worn rings/travellers, damaged aprons, misaligned guides and excessive spindle speed all raise hairiness; hairiness is an early-warning gauge of ring-frame condition.
- **Post-spinning processes** — winding raises it (above); **singeing** (passing the yarn through a flame/gas burner) burns hairs off for premium smooth yarns; waxing and sizing change its weaving behaviour without changing the spun value.

4. Interpretation and specification

- Compare like with like: H depends on count (finer yarns of the same route generally read differently from coarse), fibre, twist and test stage. Use percentile statistics for the same count/route as benchmarks, exactly as for evenness (Topic 13).
- Watch **S3 (long hairs)** separately from H for weaving and pilling-critical goods: two yarns of equal H but different S3 behave differently on the loom and in wear.
- Report test length and speed, number of packages, and the stage (cop/cone). A single-package H value is a sample of one.
- Hairiness variation (CV of H between packages) is itself diagnostic — one bad spindle position announces itself in hairiness before it shows in strength.

Glossary

Hairiness — fibre ends and loops protruding from the yarn body. **H index** — total protruding fibre length (cm) per cm of yarn in the standard measuring field. **S3** — count of hairs ≥ 3 mm long per stated test length. **Compact spinning** — ring spinning with a condensing zone that narrows the fibre strand before twisting, sharply reducing hairiness. **Singeing** — burning protruding hairs from yarn by passage through a flame. **Fly** — short fibres released into the air during processing.

Key points

• Hairiness = escaped fibre ends and loops; moderate levels soften fabric, excess causes pilling, clinging warps and fly.
• Photoelectric testers report H (total hair length per unit length), counts by length class, and S3 for the harmful long hairs.
• Short fibre content is the material cause; combing, twist and compact spinning are the cures; winding

re-creates the problem.

• Always state stage, count and route with an H value — comparisons without them are invalid.

Practice

1. A yarn's H rises from 4.8 at the cop to 5.9 on the cone. Find the percentage increase and name the stage responsible.
2. Two Ne 30 yarns both read H = 5.0; yarn A has S3 three times that of yarn B. Which do you choose for a warp, and why?
3. Give three independent changes — one raw-material, one setting, one machine-system — that each reduce hairiness, with the mechanism of each.
4. Why is a hairiness figure measured on yarn before singeing not a valid specification for the singed yarn made from it?

Answer key

1. Increase = (5.9 − 4.8)/4.8 × 100 = **22.9 %**, caused by winding (guide friction, tension and package abrasion raising and loosening hairs).
2. Yarn B. Pilling and warp clinging are driven by the long hairs counted in S3; at equal total hairiness (H), B carries far fewer long hairs, so it will weave cleaner and pill less.
3. Raw material: comb the cotton (or lower SFC by better mixing) — fewer short fibres means fewer free ends to protrude. Setting: raise twist multiplier — the tighter helix binds ends into the body. System: change to compact spinning — condensing the strand before twist insertion traps nearly all ends inside the yarn.
4. Singeing physically removes the hairs the specification counts. The singed yarn's H is a different, lower quantity produced by a later process; a specification must be measured on the product in the state in which it is sold and used.